

n8n Production Playbook

Deploy, Automate, Scale

ErgeneAI

Chapter 1: Production Deployment

The n8n Stack

n8n is the backbone of modern AI automation. This chapter covers production-grade deployment with Docker, Traefik, and auto-scaling.

Recommended Architecture

- Docker Compose with Traefik reverse proxy
- PostgreSQL for persistent workflow storage
- Redis for queue management
- Volume mounts for workflow persistence

Environment Variables

Critical production variables to configure:

- N8N_ENCRYPTION_KEY: 32-character random string
- DB_TYPE: postgresdb
- DB_POSTGRESDB_DATABASE: production
- N8N_HOST: your-domain.com
- WEBHOOK_URL: https://your-domain.com
- N8N_PROTOCOL: https

Security Checklist

- Disable user registration after setup
- Enable authentication
- Use HTTPS with valid SSL certificates
- Restrict webhook IPs where possible
- Regular backups of workflows

Chapter 2: Error Handling & Recovery

Defensive Patterns

Every production workflow needs error boundaries. n8n provides error workflows and retry mechanisms.

Error Workflow Setup

1. Create a workflow with "Error Trigger" node
2. Configure it to catch all errors
3. Send notifications to Telegram/Slack
4. Log errors to file or database

Retry Strategies

- Network calls: 3 retries with exponential backoff
- API calls: 2 retries, check idempotency
- File operations: 1 retry, check file existence
- Database operations: immediate retry, check constraints

Recovery Procedures

- Error classification: transient vs permanent
- Transient errors: auto-retry
- Permanent errors: alert + pause workflow
- Dead letter queue for failed items

Chapter 3: Webhook Security

Webhook Authentication

Protecting webhooks is critical for production systems.

Security Methods

1. Header-based auth: Custom header with HMAC signature
2. IP whitelisting: Restrict to known IP ranges
3. Token validation: JWT or API key in query params
4. Request verification: Validate payload signature

Best Practices

- Never expose internal webhooks publicly
- Use n8n's credential system for secrets
- Rotate webhook URLs periodically
- Log all webhook requests for audit
- Rate limit incoming webhooks

Chapter 4: AI Agent Workflows

Building Agent Nodes

n8n's HTTP Request node can integrate with any AI API. This chapter covers production AI agent patterns.

Agent Architecture

1. Trigger: Webhook, schedule, or event
2. Context gathering: Database, API calls, file reading
3. AI processing: OpenAI/Claude API with structured output
4. Action execution: Tool calls, API mutations
5. Response delivery: Telegram, email, webhook

Memory Management

- Short-term: n8n workflow variables
- Long-term: PostgreSQL with vector embeddings
- Session: Redis with TTL

Structured Output

Always use JSON mode or function calling for consistent AI output. Parse with Code nodes before next steps.

Cost Optimization

- Cache frequent queries
- Use smaller models for routine tasks
- Batch requests where possible
- Set max_tokens limits

Chapter 5: Telegram Integration

Telegram Bot Architecture

Production Telegram bots handle thousands of messages daily.

Setup

1. Create bot via @BotFather
2. Store token in n8n credentials
3. Set up webhook URL
4. Implement command routing

Scaling Patterns

- Rate limiting: max 30 messages/sec
- Queue management for slow operations
- Parallel processing for independent tasks
- Session management per user

Command Structure

/start -> Welcome + onboarding

/help -> Available commands

/status -> System health

/custom -> Workflow-specific commands

Chapter 6: Scheduling Patterns

Cron-Based Automation

n8n's Schedule trigger supports cron expressions for time-based automation.

Common Patterns

- Daily digests: 0 8 * * *
- Weekly reports: 0 9 * * 1
- Hourly checks: 0 * * * *
- Custom intervals: */30 * * * * (every 30 min)

Timezone Management

Always set timezone explicitly in workflow settings.

Use UTC for scheduling, convert for display.

Idempotency

Scheduled workflows must handle:

- Duplicate executions
- Missed intervals
- Daylight saving time shifts

Chapter 7: Secrets Management

Production Secrets

Never hardcode credentials. Use n8n's credential system.

Secret Types

- API Keys: Store as n8n credentials
- Database URLs: Environment variables
- Encryption keys: Docker secrets
- OAuth tokens: n8n OAuth2 integration

Rotation Policy

- API keys: Rotate every 90 days
- Database passwords: Every 180 days
- Webhook tokens: On security incident
- Access tokens: On employee departure

Chapter 8: Monitoring & Maintenance

Production Monitoring

Monitor workflow health, execution times, and error rates.

Metrics to Track

- Successful/failed executions
- Average execution time
- Queue depth
- API response times
- Error distribution

Alerting

Set up alerts for:

- Consecutive failures (3+)
- Execution time > 2x average
- Workflow stuck in loop
- Memory/CPU thresholds

Maintenance Tasks

- Weekly: Review error logs
- Monthly: Update dependencies
- Quarterly: Security audit
- Annually: Architecture review